

Data Analyst Capstone – Data Jobs

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OUTLINE



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EXECUTIVE SUMMARY



- Compensation generally increases with both age and experience.
- Full-time employees earn the most.
- Job satisfaction tends to increase with years of experience, but interestingly there is only a weak correlation with compensation and satisfaction.
- Hybrid is the most preferred work arrangement.
- JS, Python, and SQL are at the top of both "have worked with" and "want to work with."
- The largest share of respondents are:
 - 25-34-year-olds
 - Full-time employees
 - Residents of the US, India, Germany, and the UK



INTRODUCTION



- **Purpose:** I analyzed Stack Overflow survey data to find key insights about developers.
- **Target Audience:** These findings are for people who want insights into key findings from developer data.
- **Value of the Analysis:** By analyzing the survey data, this report identifies key trends, preferences, and satisfaction that can inform business strategies, priorities, and decisions.

METHODOLOGY

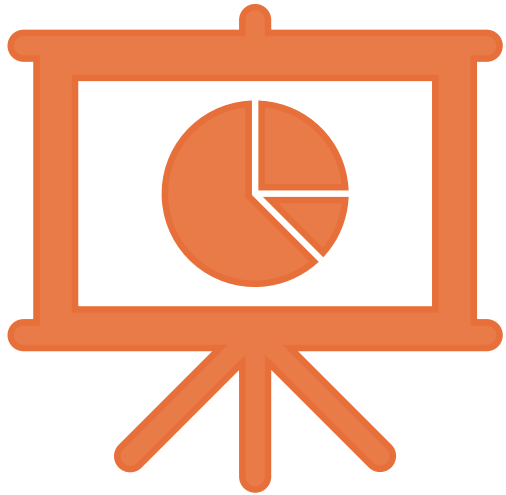
Data sources, Data collection & Data wrangling steps



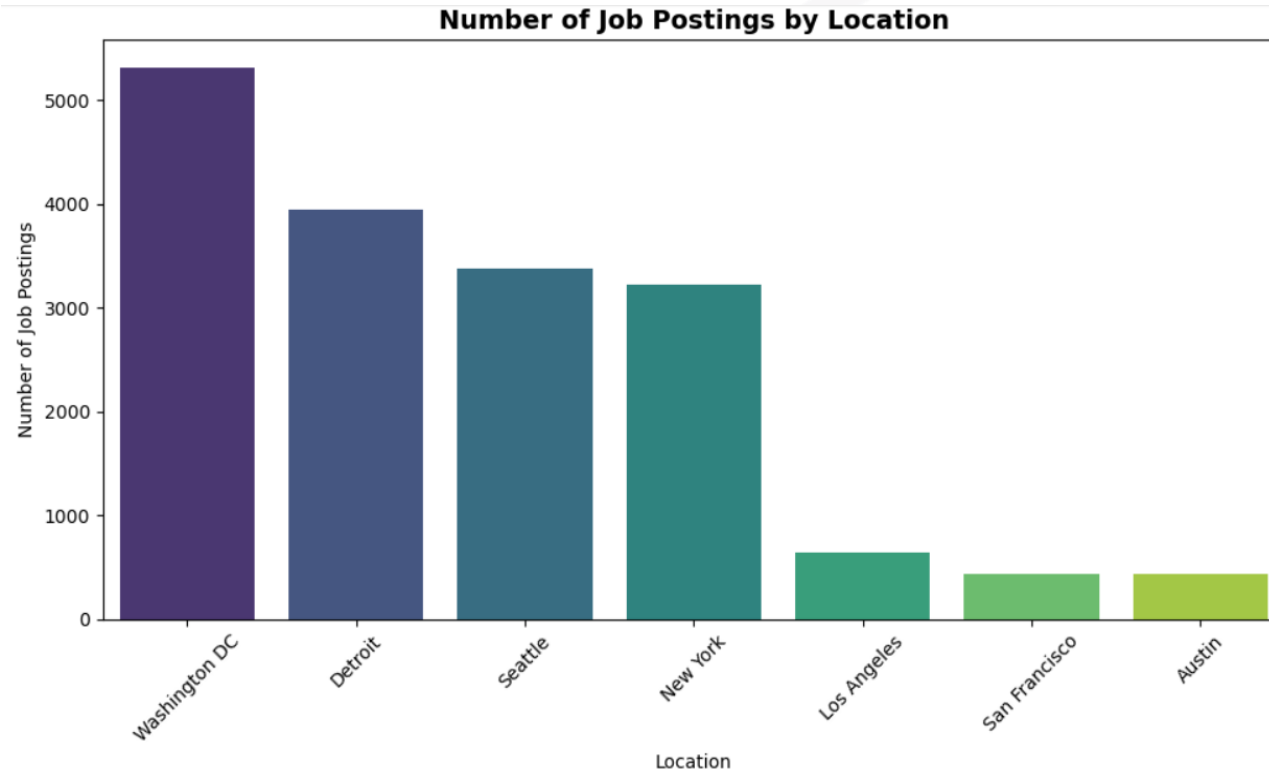
- Data Sources:
 - Popular Programming Language Data
 - Web scraped from IBM and analyzed with Python
 - Programmer Job Data
 - Web scraped from IBM and analyzed with Python
 - Stack Overflow Survey Data
 - Downloaded as CSV and uploaded to Google Looker Studio



Results



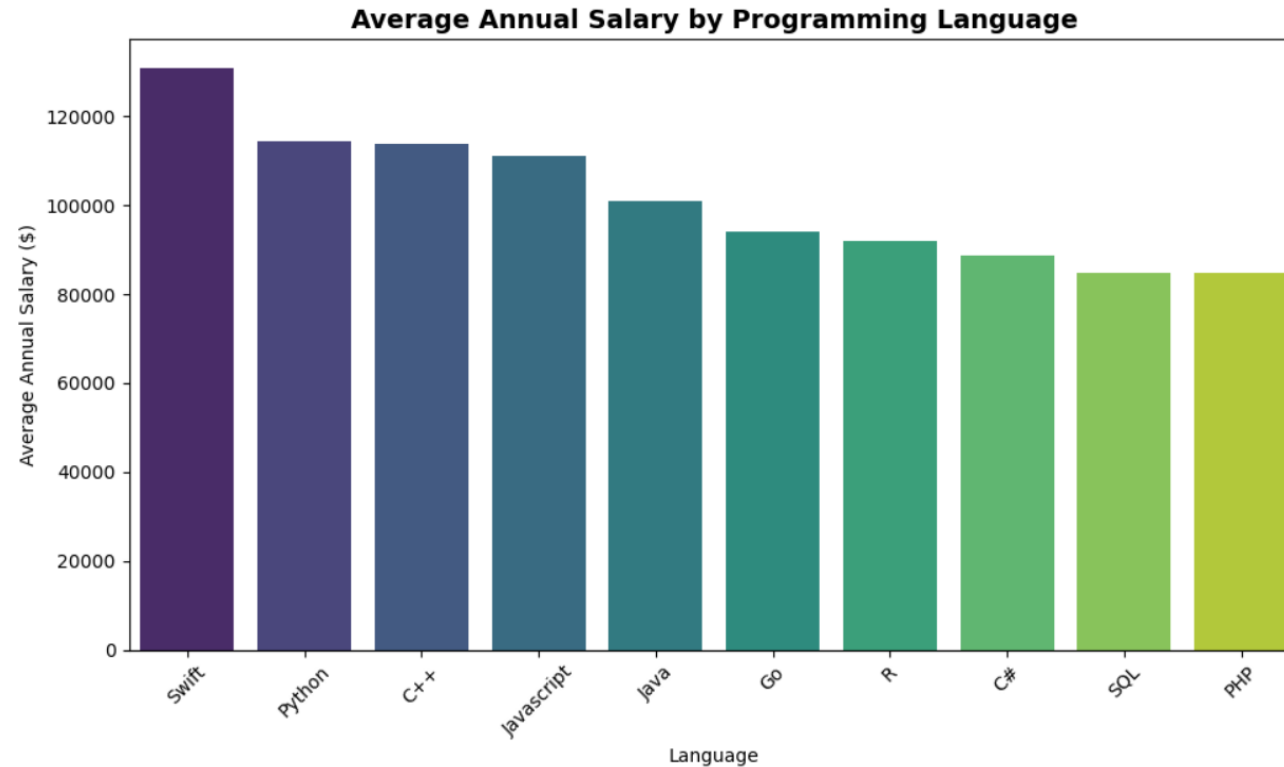
JOB POSTINGS



The Job Postings data shows that the number of job postings in this dataset vary greatly by the location, with the most amount of job postings being in Washington DC with over 5000 and the least amount being in Austin with under 500.



POPULAR LANGUAGES



The Programming Languages data shows a spread of Average Annual Salary with Swift being the most of over \$130,000 and PHP and SQL being the least with less than \$85,000 each.

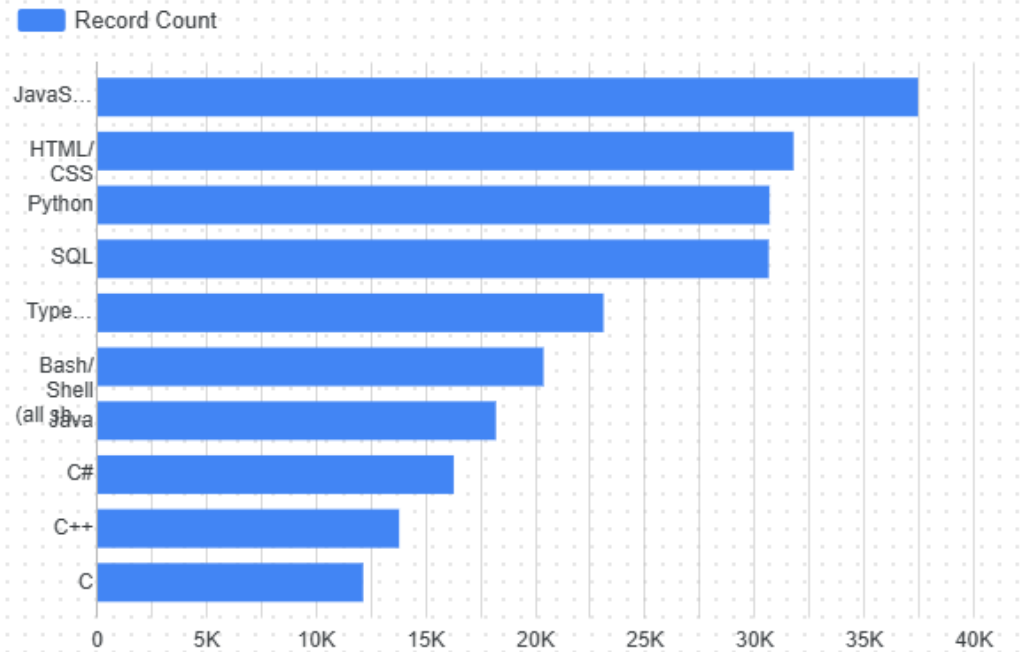


PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

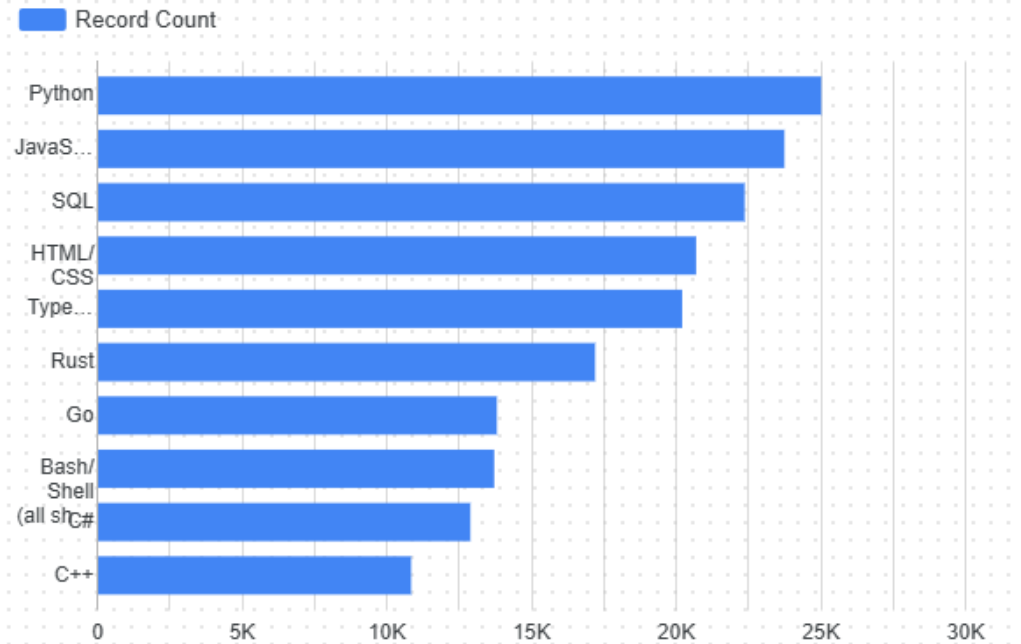
Current Year

Current Languages Used - Bar Chart



Next Year

Desired Languages - Bar Chart



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- The top 5 for both are the same for both, but the order is switched.
- Rust and Go appears on "Desired Languages" but not "Current Languages."
- Java is #7 on "Current Languages" but doesn't even appear in the top 10 of "Desired Languages."

Implications

- The top 5 for both lists having the same languages shows how those languages are currently at the core of programming.
- Python just up from #3 on "Current" to #1 on "Desired" showing how even more people want to work with Python.
- Go(2012) and Rust(2015) are newer language that are growing in implementation and popularity from people who are getting away from the dated Java(1996).



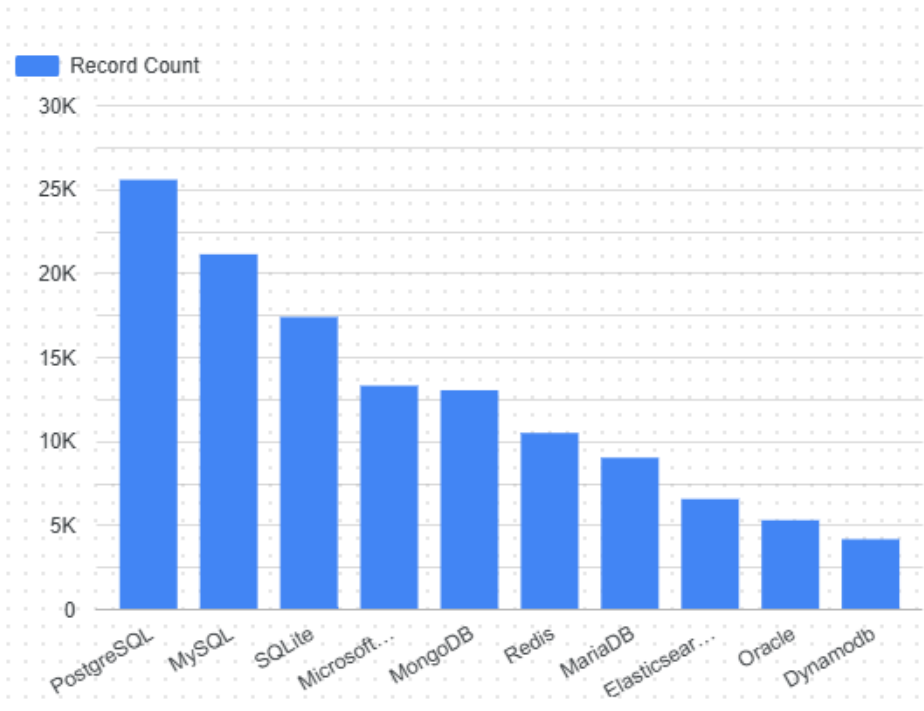
DATABASE TRENDS

Summarize key trends shown in the charts.

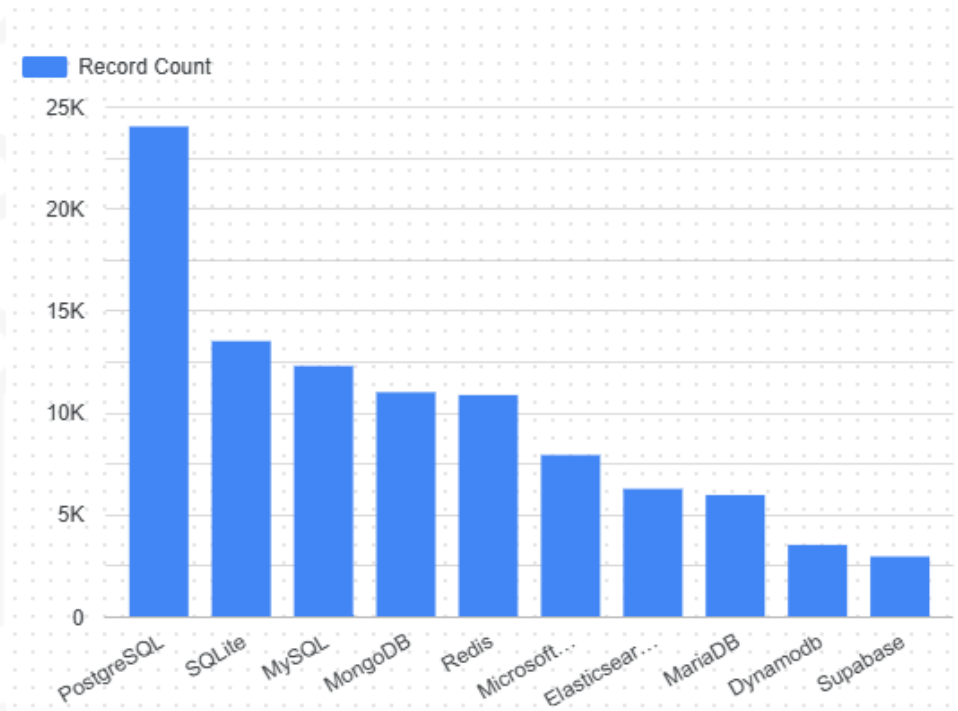
Current Year

Next Year

Current Databases Used - Column Chart



Desired Databases - Column Chart



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- The "Current" and Desired" Databases are very stable between the two.
- PostgreSQL is already at the top of "Current" but it is still a clear #1 of "Desired."

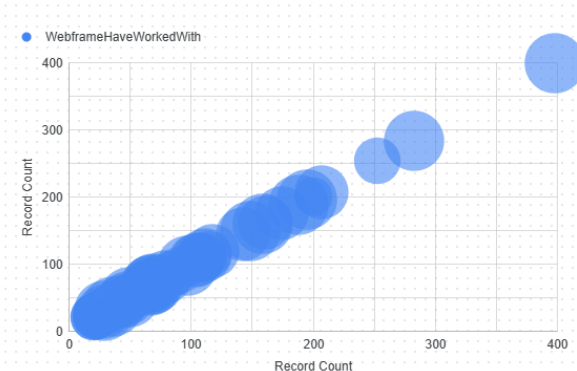
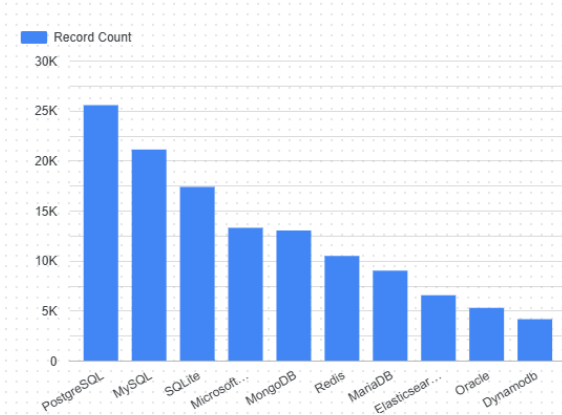
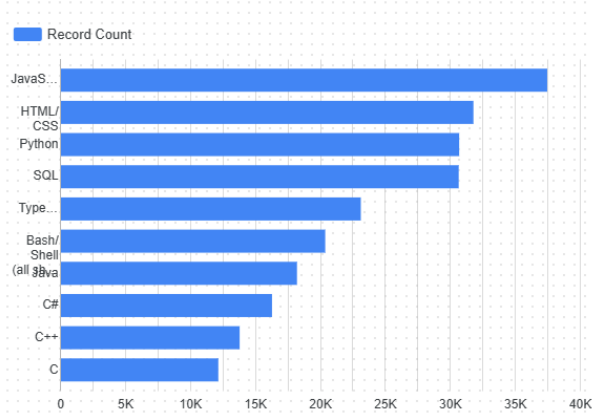
Implications

- The databases are highly stable compared to Programming Languages. This might be due to the headaches that can come from migrating Databases.
- PostgreSQL has essentially become the default choice for databases as PostgreSQL can do virtually everything natively, making it ideal for consolidating data stacks making it the clear choice for the rise in AI.

DASHBOARD



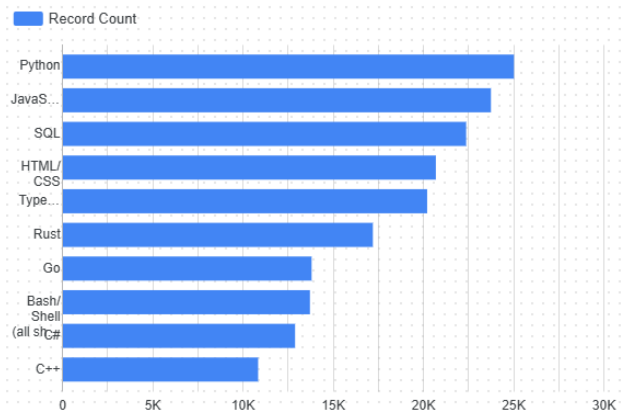
DASHBOARD TAB 1: Current Technology Usage



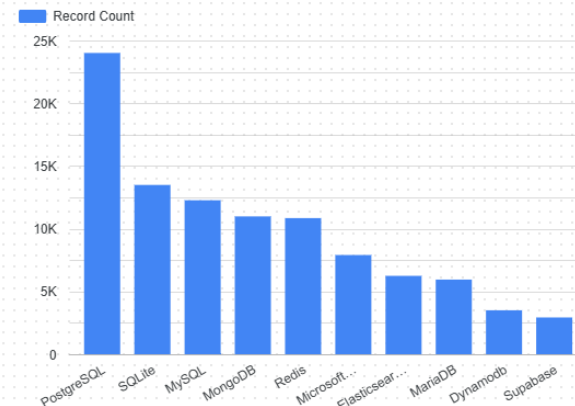
- JavaScript, HTML/CSS, Python, and SQL have a clear lead of the Current languages.
- PostgreSQL, MySQL, SQLite lead the current databases.
- AWS has a clear lead of current platforms followed by Microsoft Azure and Google Cloud.

DASHBOARD TAB 2: Future Technology Trends

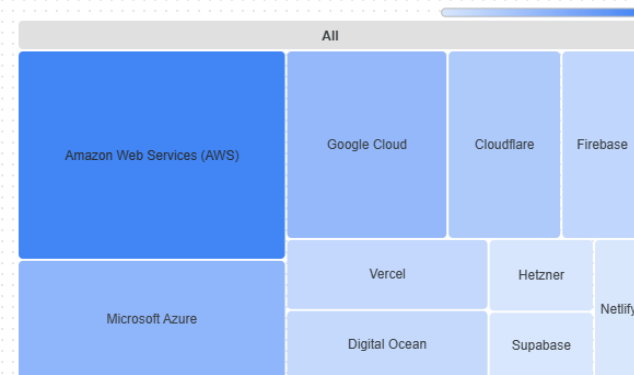
Desired Languages - Bar Chart



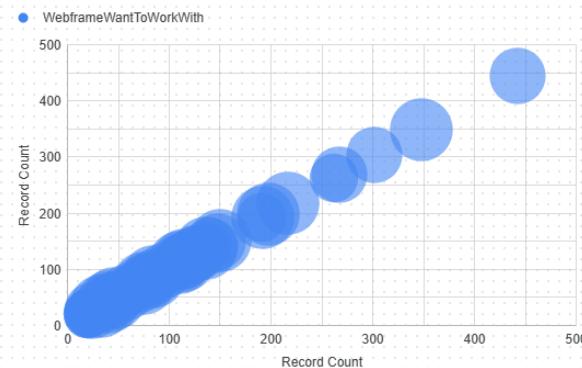
Desired Databases - Column Chart



Desired Platform - Tree Map

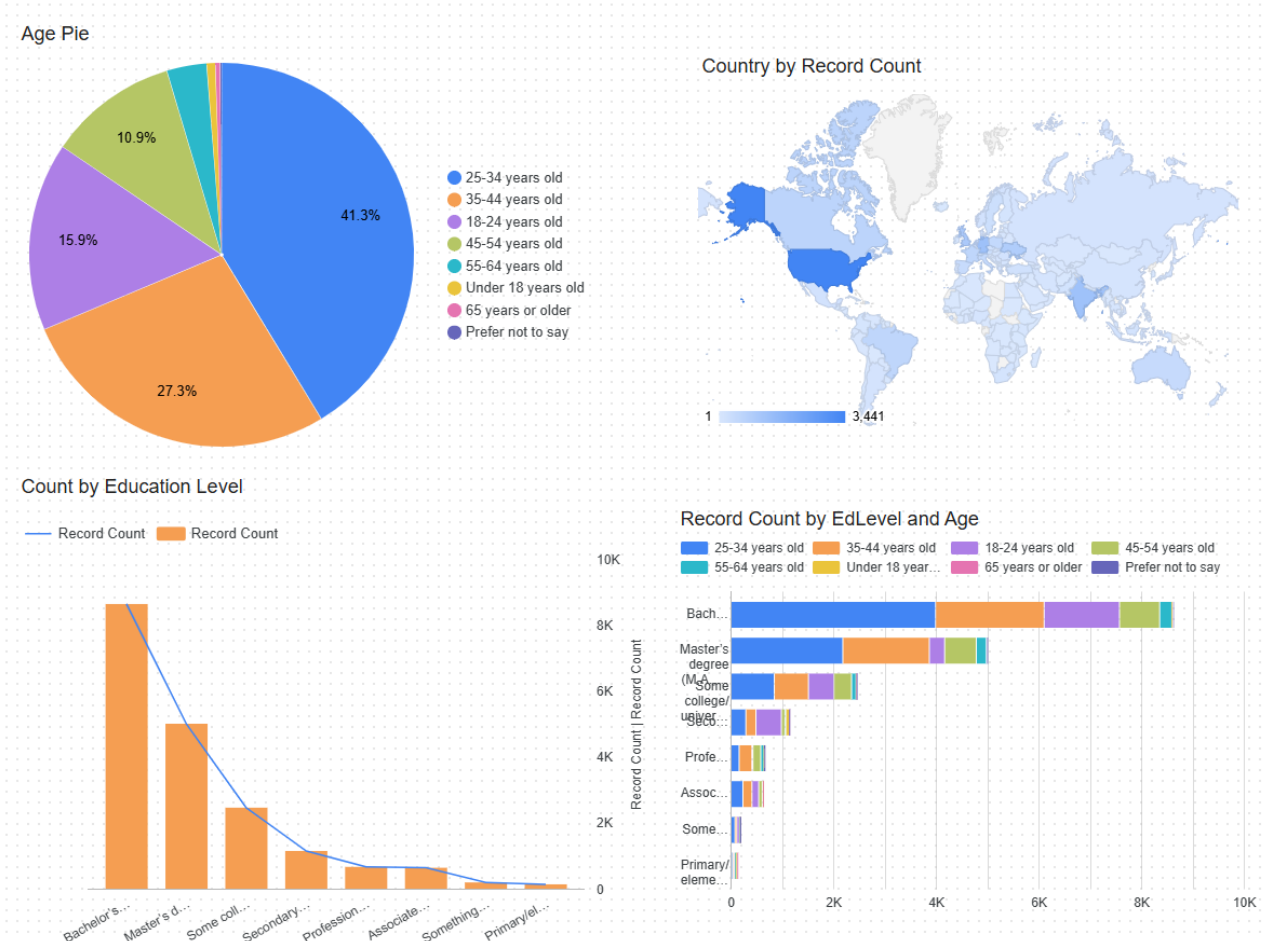


Desired Framework - Scatter Bubble



- Python goes up to the first most desired language followed by JavaScript, SQL, and HTML/CSS.
- PostgreSQL has a clear lead for most desired database, followed by SQLite and MySQL.
- AWS also has a clear lead of the desired platform and is still followed by Google Cloud and Microsoft Azure.

DASHBOARD TAB 3: Demographics



- The 25-34-year-old group has the largest share of programmers with 41.3%, followed by the 35-44-year-old group with 27.3%.
- The US has a clear lead of programmers, followed by the UK, Germany, and India.
- Bachelor's Degree is a standout for education level of programmers followed by Master's Degree.
- The Stacked bar chart roughly shows what you would expect when combining Ed Level with Age except some underrepresentation in certain areas due to higher Ed Levels requiring time.



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript is the #1 Current language, but Python is the #1 Desired language.
- The database ecosystem is stable with a lower change rate but there is a high interest in PostgreSQL.
- The ratio of Bachelor's to Master's degrees for programmers is around 1.7:1 and the national total is 2.4:1.

Implications

- Hirers and managers can expect a change in interest toward Python.
- The desire for PostgreSQL is likely due to how it works well with AI, causing companies to offer a premium for quick hires.
- This implies that there is a higher demand of people with Master's degrees in Programming.



CONCLUSION



- Job Postings for programmers can vary significantly based on location
- Salary varies heavily based on supply and demand; Swift, Python, and C++ lead in US salary.
- Desire:
 - Python is the most desired programming language
 - PostgreSQL is the most desired database
 - AWS is the most desired Platform
- Standout Demographics:
 - The 25-34-year-old group has the largest representation
 - US Citizens has the largest representation
 - People with Bachelor's degrees has the largest representation



APPENDIX

